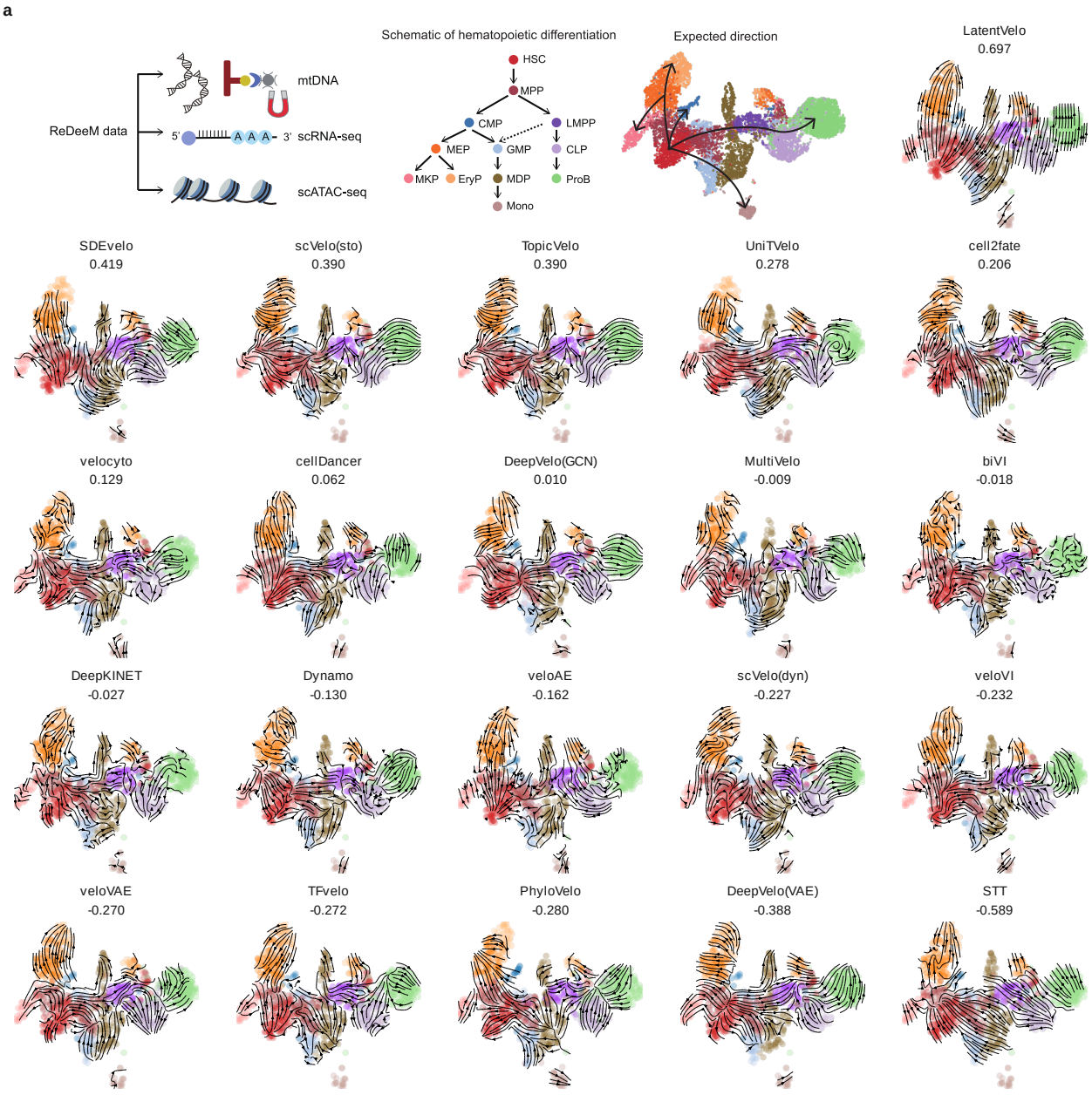
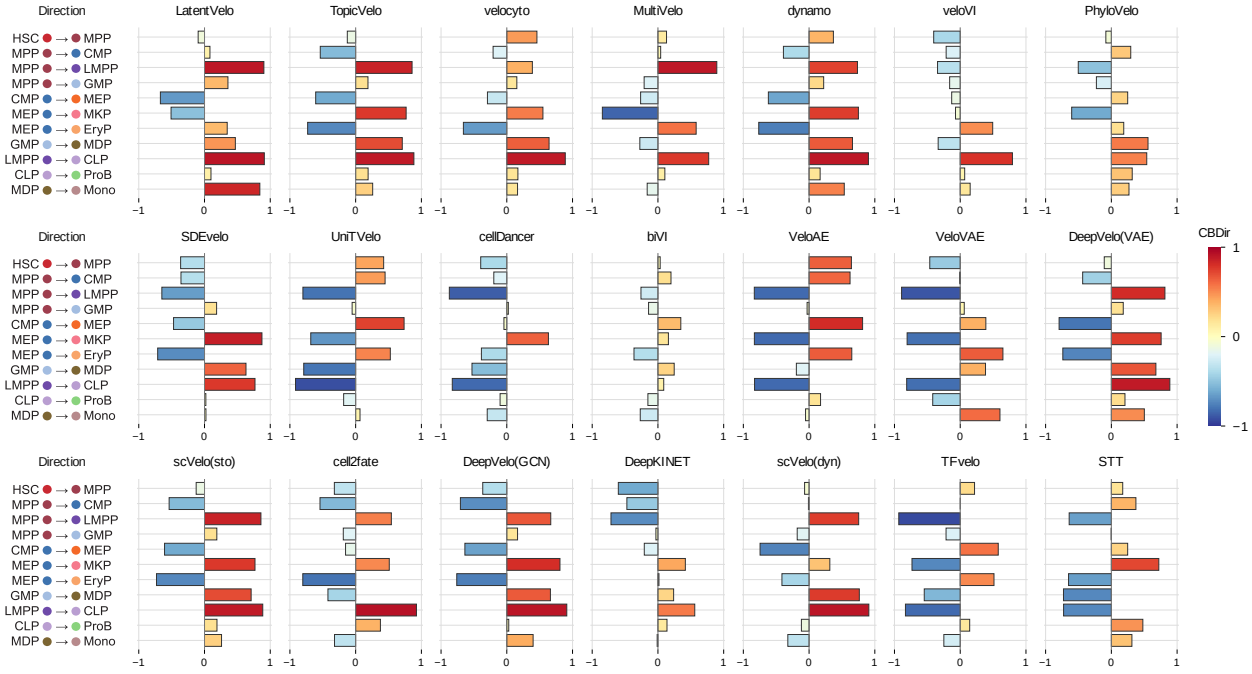

a
ReDeeM data
mtDNA
scRNA-seq
scATAC-seq
Schematic of hematopoietic differentiation
HSC
MPP
CMP
LMPP
MEP
GMP
CLP
MKP
EryP
MDP
ProB
Mono
Expected direction
LatentVelo
0.697
SDEvelo
0.419
scVelo(sto)
0.390
TopicVelo
0.390
UniTVelo
0.278
cell2fate
0.206
velocyto
0.129
cellDancer
0.062
DeepVelo(GCN)
0.010
MultiVelo
-0.009
biVI
-0.018
DeepKINET
-0.027
Dynamo
-0.130
veloAE
-0.162
scVelo(dyn)
-0.227
veloVI
-0.232
veloVAE
-0.270
TFvelo
-0.272
PhyloVelo
-0.280
DeepVelo(VAE)
-0.388
STT
-0.589


b
Direction
LatentVelo
TopicVelo
velocyto
MultiVelo
dynamo
veloVI
PhyloVelo
HSC → MPP
MPP → CMP
MPP → LMPP
MPP → GMP
CMP → MEP
MEP → MKP
MEP → EryP
GMP → MDP
LMPP → CLP
CLP → ProB
MDP → Mono
SDEvelo
UniTVelo
cellDancer
biVI
VeloAE
VeloVAE
DeepVelo(VAE)
CBDir
scVelo(sto)
cell2fate
DeepVelo(GCN)
DeepKINET
scVelo(dyn)
TFvelo
STT